

1			Q1: if consistent "0.8" incorrect or 1/8, 7/8 or 0.02 allow M marks in ii , iii & 1st M1 in i
i	Binomial stated 0.9437 - 0.7969 or ${}^8C^3 \times 0.2^3 \times 0.8^5 = 0.147$ (3 sfs)	M1 M1 A1 3	or implied by use of tables or ${}^8C^3$ or $0.2^a \times 0.8^b$ (a+b= 8)
ii	$1 - 0.7969 = 0.203$ (3 sf)	M1 A1 2	allow 1- 0.9437 or 0.056(3) or equiv using formula
iii	8×0.2 oe 1.6	M1 A1 2	$8 \times 0.2 = 2$ M1A0 1.6 : 8 or 1/1.6 M0A0
Total		7	
2	first two d's ± 1 Ed2 attempted (=2) 1- 6 x "2" 7(72 1) 27/28 or 0.964 (3 sfs)	B1 M1 M1dep A1	S or Syy 28 B1 $S_{xy} = 27$ B1 S_{xy} N(S_{xx} S_{yy}) M1 dep B1 1234567 & 1276543 (ans 2/7): MR, lose A1
Total		4	
31	x independent or controlled or changed Value of y was measured for each x x not dependent	B1 1	Allow Water affects yield, or yield is dependent or yield not control water supply Not just y is dependent Not x goes up in equal intervals Not x is fixed
ii	(line given by) minimum sum of squas	B1 B1 2	B1 for "minimum" or "least squares" with inadequate or no explanation
iii	$S_{xx} = 17.5$ or 2.92 $S_{yy} = 41.3$ or 6.89 $S_{xy} = 25$ or 4.17 $r = S_{xy} / \sqrt{(S_{xx} S_{yy})} =$ 0.930 (3 sf)	B1 M1 A1 3	or 91 - 216 or 394 - 462/6 B1 for any one or 186-21×46 6 dep B1 0.929 or 0.93 with or without wking B1M1A0 SC incorrect n: max B1M1A0
iv	Near 1 or lg, high, strong, good corr'n or relnship oe Close to st line or line good fit	B1ft B1 2	r small: allow little (or no) corr'n oe Not line accurate. Not fits trend
Total		8	